Computer Networks
Fall 2024/25
SOLUTION
Exercise-2-solution

Problem 1 (20 points).

Question	response time (in msec)
1	**6601.1 msec**
2	**1200.2 (did NOT accept 1201.1) mesc**
3	**3601.1 msec**
4	**901.1 msec**

Problem 2 (20 points).
RFC – check.

Problem 3 (20 points).
1 – it should be possible.
2 – there are different possible answers, as long as the student tries to solve it is ok.

Problem 4 (20 points).
Check.

Problem 5 -- Lecture #5 (20 points).
1. Answer: $k = \lfloor t/5 \rfloor = 40/5 =$ **8**.
Given 40 faults (1) during the first 40/5 rounds, it is possible that there will not be even a single clean round, and (2) during at most 28/4 + 1 rounds, there must be a clean round. The result follows.
2. Answer: $n=27$, $t=7$, $s=n-t=20$.
NO. Lemma 2 still holds. In such a case, we can use Lemma 2 of the proof (it still holds!), which states that if all correct processes had the same input value before the round, they would all keep that value.
3. Answer: Yes. If the leader of the first round is byzantine, then after the first round, the value of all the processes can be 1.